

De Moivre's Theorem and Euler Form

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Learning Outcome

At the end of this lecture, you should be able to:

- Know what is De Moivre's Theorem and how to apply it.
- Express $\cos(n\theta)$ and $\sin(n\theta)$ in terms of powers of $\cos \theta$ and $\sin \theta$.
- Express powers of $\cos \theta$ and $\sin \theta$ in terms of $\cos(k\theta)$ and $\sin(k\theta)$.
- Write a complex number in the Euler form and convert a complex number in Euler form into Cartesian or Polar form. *or vice versa.*

De Moivre's Theorem

De Moivre's Theorem:

For any $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$(\cos \theta + i \sin \theta)^n = \cos \underline{n\theta} + i \sin n\theta. \quad (*)$$

Proof (by induction).

Case 1: $n=0$.

$$(\cos \theta + i \sin \theta)^0 = 1$$

$$\underbrace{\cos 0}_1 + i \underbrace{\sin 0}_0 = 1$$

So (*) is true for $n=0$.

De Moivre's Theorem

Case 2: $n \geq 1$

Base Step: When $n=1$, $(\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta$

$$= \cos 1 \cdot \theta + i \sin 1 \cdot \theta$$

So (*) is true for $n=1$.

Inductive Step: Suppose that (*) is true for $n=k \geq 1$

$$\text{When } n=k+1, (\cos \theta + i \sin \theta)^{k+1} = (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$= (\cos(k\theta) + i \sin(k\theta)) (\cos \theta + i \sin \theta)$$

$$= \cos(k\theta) \cos \theta + i \cos(k\theta) \sin \theta + i \sin(k\theta) \cos \theta + i^2 \sin(k\theta) \sin \theta$$

$$\cos A \cos B - \sin A \sin B = \cos(A+B)$$

$$\cos A \sin B + \sin A \cos B = \sin(A+B)$$

$$= \cos(k\theta) \cos \theta - \sin(k\theta) \sin \theta$$

$$+ i (\cos(k\theta) \sin \theta + \sin(k\theta) \cos \theta)$$

$$= \cos(k\theta + \theta) + i \sin(k\theta + \theta)$$

$$= \cos[(k+1)\theta] + i \sin[(k+1)\theta]$$

So (*) is true for $n = k+1$.

By mathematical induction, (*) is true for all integer $n \geq 1$.

Case 3: $n \leq -1$

Let $m = -n$, then $m \geq 1$

$$\frac{(\cos \theta + i \sin \theta)^n}{1} = \frac{1}{(\cos \theta + i \sin \theta)^m}$$

$$= \frac{1}{\cos(m\theta) + i \sin(m\theta)}, \text{ from case 2.}$$

$$= \frac{1}{\cos(m\theta) + i \sin(m\theta)} \cdot \frac{\cos(m\theta) - i \sin(m\theta)}{\cos(m\theta) - i \sin(m\theta)}$$

$$= \frac{\cos(m\theta) - i \sin(m\theta)}{\cos^2(m\theta) + \sin^2(m\theta)} = 1$$

$$= \cos(m\theta) - i \sin(m\theta)$$

$$= \cos(-n\theta) - i \sin(-n\theta)$$

$$= \cos(n\theta) - (-i \sin(n\theta))$$

$$= \underline{\cos(n\theta) + i \sin(n\theta)}$$

So (*) is true for all integer $n \leq -1$

Combining all cases, (*) is true for all integer n .

From Calculus,
 $\cos x$ is an even
 function,

$$\cos(-x) = \cos x$$

$\sin x$ is an odd
 function,

$$\sin(-x) = -\sin x$$

De Moivre's Theorem

Let $z = r(\cos \theta + i \sin \theta)$, and n be a positive integer. Then

$$z^n = [r(\cos \theta + i \sin \theta)]^n = r^n (\cos \theta + i \sin \theta)^n = r^n [\cos(n\theta) + i \sin(n\theta)]$$

$$z^{-n} = (r \cos \theta + i \sin \theta)^{-n} = r^{-n} (\cos \theta + i \sin \theta)^{-n} \\ = \frac{1}{r^n} [\cos(-n\theta) + i \sin(-n\theta)]$$

$$\frac{1}{z^n} = \frac{1}{r^n} [\cos(n\theta) - i \sin(n\theta)]$$

Example:

Let $z = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$.

$$z^{12} = [2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})]^{12} = 2^{12} (\cos(12(\frac{2\pi}{3})) + i \sin(12(\frac{2\pi}{3}))) \\ = 2^{12} (\cos 8\pi + i \sin 8\pi) = 2^{12}$$

Application of De Moivre's Theorem

Find $(-1 + i)^{202}$.

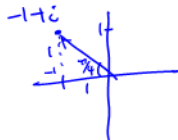
First, we convert $(-1 + i)$ into polar form.

$$r = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$$

$$\theta = \text{Arg}(-1 + i) = \pi - \arctan\left(\frac{1}{1}\right)$$

$$= \pi - \frac{\pi}{4}$$

$$= \frac{3\pi}{4}$$



$$\text{So } -1 + i = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$(-1 + i)^{202} = (\sqrt{2})^{202} \left[\cos \frac{202}{2} \left(\frac{3\pi}{4} \right) + i \sin \frac{202}{2} \left(\frac{3\pi}{4} \right) \right]$$

$$= 2^{101} \left[\cos \frac{3 \cdot 202\pi}{2} + i \sin \frac{3 \cdot 202\pi}{2} \right]$$

$$= 2^{101} \left[\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right] = 2^{101} (0 + i(-1)) = -2^{101} i$$

Application of De Moivre's Theorem

Compute $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$.

$$\frac{1+\sqrt{3}i}{1-\sqrt{3}i} = \frac{1+\sqrt{3}i}{1-\sqrt{3}i} \cdot \frac{1+\sqrt{3}i}{1+\sqrt{3}i} = \frac{(1+\sqrt{3}i)^2}{4}$$

$$\frac{1+2\sqrt{3}i+3i^2}{4} = \frac{-2+2\sqrt{3}i}{4}$$

↓
polar form
↓
power of 10.

Let $w = 1+\sqrt{3}i$ (1st quadrant)

$$|w| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\text{Arg}(w) = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

$$w = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

$$\left[\frac{(1+\sqrt{3}i)^2}{4}\right]^{10} = \frac{[2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})]^{10}}{4^{10}} = \frac{2^{10}(\cos \frac{20\pi}{3} + i \sin \frac{20\pi}{3})}{4^{10}}$$
$$= \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$
$$= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$



Application of De Moivre's Theorem

Express $\cos n\theta$ and $\sin n\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.

(a) Find formulas for $\cos(4\theta)$ and $\sin(4\theta)$ in terms of powers of $\cos \theta$ and $\sin \theta$.

(b) Hence or otherwise prove that $\cos(4\theta) + 8\cos^2 \theta \geq 1$ for all $\theta \in \mathbb{R}$.

(a) From De Moivre's Theorem,

$$\begin{aligned}\cos(4\theta) + i\sin(4\theta) &= (\cos \theta + i\sin \theta)^4 \quad \text{Binomial Theorem} \\ &= \cos^4 \theta + 4\cos^3 \theta (i\sin \theta) \\ &\quad + 6\cos^2 \theta (i\sin \theta)^2 + 4\cos \theta (i\sin \theta)^3 \\ &\quad + (i\sin \theta)^4\end{aligned}$$

Comparing the real and imaginary part,

$$\begin{aligned}\cos(4\theta) &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta \\ &\quad + \sin^4 \theta,\end{aligned}$$

$$\begin{aligned}\sin(4\theta) &= 4\cos^3 \theta \sin \theta \\ &\quad - 4\cos \theta \sin^3 \theta.\end{aligned}$$

$$\begin{aligned}&= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta \\ &\quad + i(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta)\end{aligned}$$

Application of De Moivre's Theorem

Express $\cos n\theta$ and $\sin n\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.

*a) Find formulas for $\cos(4\theta)$ and $\sin(4\theta)$ in terms of powers of $\cos \theta$ and $\sin \theta$.

(b) Hence or otherwise prove that $\cos(4\theta) + 8\cos^2 \theta \geq 1$ for all $\theta \in \mathbb{R}$.

$$\text{b)} \quad \cos 4\theta + 8\cos^2 \theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta + 8\cos^2 \theta$$

$$= \cos^4 \theta - 6\cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta) + 8\cos^2 \theta$$

$$= \cos^4 \theta - 6\cos^2 \theta + 6\cos^4 \theta + 1 - 2\cos^2 \theta + \cos^4 \theta + 8\cos^2 \theta$$

$$= 8\cos^4 \theta + 1$$

$$\geq 1$$

Since, $\cos^4 \theta = (\cos^2 \theta)^2 \geq 0$
for all $\theta \in \mathbb{R}$.

Application of De Moivre's Theorem

Express powers of $\cos \theta$ and $\sin \theta$ in terms of cosines and sines of multiples of θ .

Let $z = \cos \theta + i \sin \theta$ and n is a positive integer. Show that

(i) $2 \cos n\theta = z^n + z^{-n}$ ✓

(ii) $2i \sin n\theta = z^n - z^{-n}$ ✓

Using (i) and (ii), express $\sin^4 \theta$ in term of cosines of multiples of θ cos(nθ)

By De Moivre's theorem,

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)) \quad \text{--- ①}$$

$$z^{-n} = r^{-n} (\cos(n\theta) - i \sin(n\theta)) \quad \text{--- ②}$$

$$\text{①} + \text{②}, \quad z^n + z^{-n} = 2 \cos(n\theta)$$

$$\text{①} - \text{②}, \quad z^n - z^{-n} = 2i \sin(n\theta)$$

$$z^n + z^{-n} = 2 \cos(n\theta) \Rightarrow \cos(n\theta) = \frac{z^n + z^{-n}}{2}$$

$$z^n - z^{-n} = 2i \sin(n\theta) \Rightarrow \sin(n\theta) = \frac{z^n - z^{-n}}{2i}$$

$$n=1, \cos \theta = \frac{z + z^{-1}}{2}; \quad \sin \theta = \frac{z - z^{-1}}{2i}$$

$$\sin^4 \theta = \left(\frac{z - z^{-1}}{2i} \right)^4 = \frac{(z - z^{-1})^4}{(2i)^4}$$

binomial theorem

$$(2i)^4 = 16$$

$$= \frac{1}{16} \left(z^4 + 4 \overset{-2}{z} \overset{2}{(-z^{-1})} + 6 \overset{-2}{z} \overset{2}{(-z^{-1})^2} + 4 \overset{-2}{z} \overset{2}{(-z^{-1})^3} + (-z^{-1})^4 \right)$$

$$= \frac{1}{16} (z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4})$$

$$= \frac{1}{16} \left[\underbrace{(z^4 + z^{-4})}_{2 \cos 4\theta} - 4 \underbrace{(z^2 + z^{-2})}_{2 \cos 2\theta} + 6 \right]$$

$$= \frac{1}{16} [2 \cos 4\theta - 8 \cos 2\theta + 6]$$

$$= \frac{1}{8} (\cos 4\theta - 4 \cos 2\theta + 3)$$

Exponential Form

Cartesian form: $z = x + yi$, $x, y \in \mathbb{R}$

$x: \operatorname{Re}(z)$
 $y: \operatorname{Im}(z)$

Polar form: $z = r(\cos \theta + i \sin \theta)$, where $r = \sqrt{x^2 + y^2}$, $\tan \theta = \frac{y}{x}$,
 $x = r \cos \theta$ and $y = r \sin \theta$.

$r: \text{modulus}$
 $\theta: \text{argument / principal argument.}$

Some infinite series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

You will learn more on this in Calculus II

Exponential Form

$$e^{i\theta} = 1 + \underline{i\theta} + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots$$

$$\begin{aligned} i^2 &= -1 & i^4 &= 1 \\ i^3 &= -i & i^6 &= -i \end{aligned}$$

$$= \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right)}_{\cos(\theta)} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)}_{\sin \theta}$$

$$e^{i\theta} = \cos \theta + i \sin \theta \leftarrow \text{Euler Formula.}$$

$$z = x + yi$$

In general,

$$e^z = e^{x+yi} = e^x e^{iy} = e^x (\cos y + i \sin y) \quad \text{Euler formula}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta)$$

$$= \cos \theta - i \sin \theta$$

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta \Rightarrow \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta \Rightarrow \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

polar form: $z = r(\cos \theta + i \sin \theta)$

$$= r \left(\frac{\cancel{e^{i\theta}} + \cancel{e^{-i\theta}}}{2} + \frac{1(\cancel{e^{i\theta}} - \cancel{e^{-i\theta}})}{2i} \right)$$

$$= r \left(\frac{\cancel{1}e^{i\theta}}{\cancel{1}} \right) = \underline{r e^{i\theta}} \leftarrow \text{exponential form of } z.$$

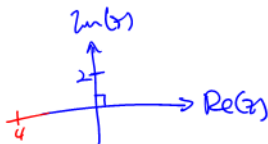
Exponential Form

Example: Express $z = 2i$ in exponential form.

$$r = 2, \text{Arg}(z) = \frac{\pi}{2}$$

$$z = re^{i\theta} = 2e^{i\pi/2}$$

$$z^2 = (2e^{i\pi/2})^2 = 2^2 (e^{i\pi/2(2)}) = 4e^{i\pi} = -4$$



Example: Express $z = 2 + 2\sqrt{3}i$ in exponential form.

$$r = \sqrt{\underbrace{(2)^2}_{-4} + \underbrace{(2\sqrt{3})^2}_{-12}}$$
$$= 4$$

$$\text{Arg}(z) = \arctan\left(\frac{2\sqrt{3}}{2}\right)$$

$$= \pi/3$$

$$z = 4e^{i\pi/3}$$

